A REPRESENTABLE MONOMORPHISM OF FIBRED CATEGORIES IS FULLY FAITHFUL ON FIBRE CATEGORIES

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Editorial note. No omission markers in-span. The faithfulness input is not carded (pure citation combination) and is declared external.

Allowed standard prerequisites (editorial).

- Faithfulness criterion. lemma-criterion-map-representable-spaces-fibred-in-groupoids (same chapter, NOT carded, declared external). Supplies faithfulness on fibre categories.
- 2-Yoneda embedding. the 2-Yoneda section of this chapter (same chapter, structural). Views objects u, v as 1-morphisms from (Sch/U).
- Monomorphism with a section is an isomorphism. Descent on Spaces, lemma descending-property-monomorphism plus the elementary section fact (cross-chapter, named). Forces F_{j o v} = U.

Source reference. Original definition 03YK: sources/stacks-algebraic/pinned-original-source.tex, source lines 1061–1061. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02ZY: sources/stacks-algebraic/pinned-original-source.tex, source lines 769–769. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 04SS: sources/stacks-algebraic/pinned-original-source.tex, source lines 386–386. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042D: focused-seeds/geometry/S9-alg-open-fibred-category-is-full/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $j : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. Assume j is representable by algebraic spaces and a monomorphism (see Definition 03YK and Descent on Spaces, Lemma 042D). Then j is fully faithful on fibre categories.*

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We have seen in Lemma 02ZY that j is faithful on fibre categories. Consider a scheme U, two objects u, v of $\mathcal{X}_U$, and an isomorphism $t : j(u) \to j(v)$ in $\mathcal{Y}_U$. We have to construct an isomorphism in $\mathcal{X}_U$ between u and v. By the 2-Yoneda lemma (see Section 04SS) we think of u, v as 1-morphisms $u, v : (Sch/U)_{fppf} \to \mathcal{X}$ and we consider the 2-fibre product

$$(Sch/U)_{fppf} \times_{j \circ v, \mathcal{Y}} \mathcal{X}.$$

By assumption this is representable by an algebraic space $F_{j \circ v}$, over U and the morphism $F_{j \circ v} \to U$ is a monomorphism. But since $(1_U, v, 1_{j(v)})$ gives a 1-morphism of $(Sch/U)_{fppf}$ into the displayed 2-fibre product, we see that $F_{j \circ v} = U$ (here we use that if $V \to U$ is a monomorphism of algebraic spaces which has a section, then $V = U$). Therefore the 1-morphism projecting to the first coordinate

$$(Sch/U)_{fppf} \times_{j \circ v, \mathcal{Y}} \mathcal{X} \to (Sch/U)_{fppf}$$

is an equivalence of fibre categories. Since $(1_U, u, t)$ and $(1_U, v, 1_{j(v)})$ give two objects in $((Sch/U)_{fppf} \times_{j \circ v, \mathcal{Y}} \mathcal{X})_U$ which have the same first coordinate, there must be a 2-morphism between them in the 2-fibre product. This is by definition a morphism $\tilde{t} : u \to v$ such that $j(\tilde{t}) = t$. $\square$